

A photograph of the entrance to The Marine Science Institute. The building is a concrete structure with a large set of stairs leading up to the entrance. Above the entrance, the text "The MARINE SCIENCE INSTITUTE" is visible, along with a circular logo featuring a globe and the letters "MSI".

The MARINE SCIENCE INSTITUTE

5 weeks at MSI
(MBRIS+MBEL)

May 29-July 4

A photograph of a dirt path leading through a lush, green forest. The path is surrounded by tall trees and dense foliage. In the foreground, there is a metal railing, possibly part of a staircase or walkway. The sky is visible through the canopy of trees.



Casmaria ponderosa (f) nodulosa
Family: Cassidae



Conus ebraeus
Family: Conidae



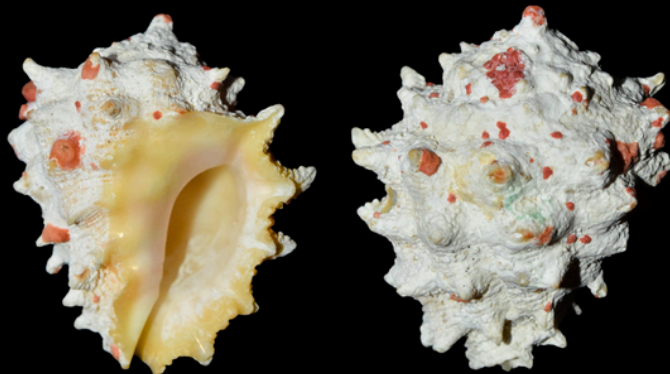
Latirolagena smaragdula
Family: Fascioliidae



Drupina grossularia
Family: Muricidae



Anadara antiquata
Family: Arcidae



Ricinella rubusidaeus
Family: Muricidae



Nassarius papillosus
Family: Nassariidae



What we did:

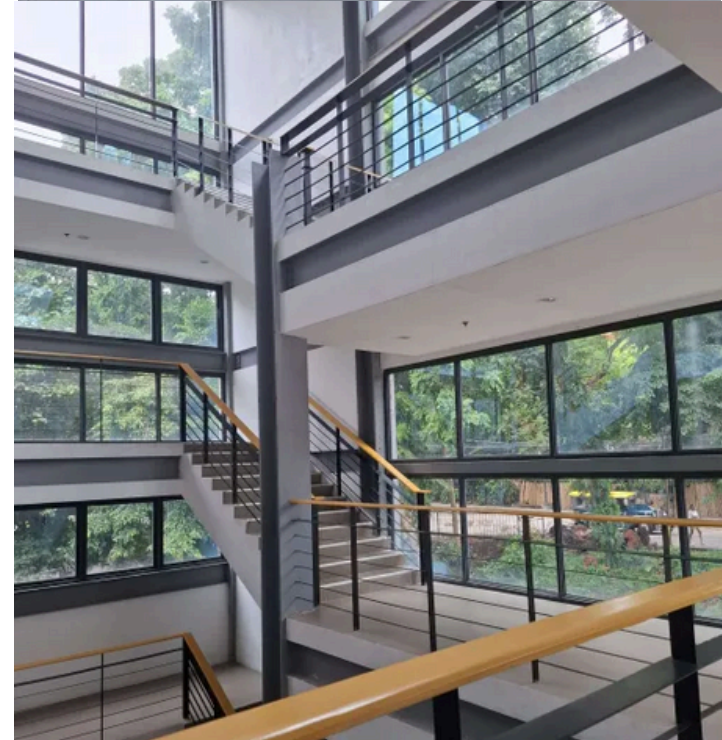
- Introduction to the lab and its specialization (invertebrates).
- Tour around MSI and MBRIS.
- Attempted to ID (at the very least to the Family level) different mollusks and some cnidarians (all dry specimen).

Insights and thoughts:

Despite only having two days, this week I was introduced to the process of classifying organisms, at least preliminarily. I got used to using dichotomous keys to identify the families of mollusks (primarily bivalves) based on the materials given to us. I realize it's important to be familiar with the technical terms for each component of an organism (e.g. umbo, pallial line) in order to have a better grasp on the descriptions. Common descriptor words used in scientific literature are also good to learn (e.g. ovate, tubular) so that it is easier to verbalize or note down unique characteristics.

Other notes unrelated to the main activity:

The facilities at MSI are very nice, despite some still being under construction or unfurnished. I only got a glimpse of them, but the sheer number of samples is amazing – and this is only for one lab.



What we did: Organized Pag-asa Invertebrate Samples

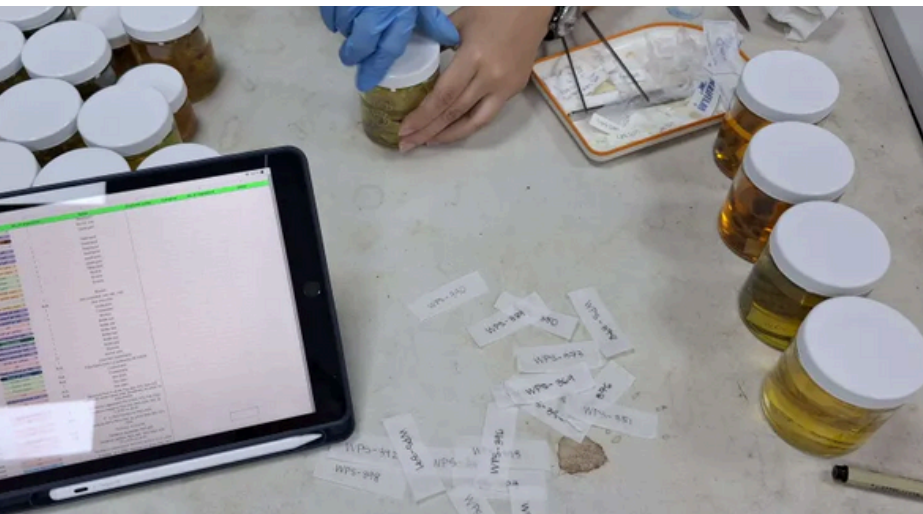
- Made a database for all of the samples separated by container. The information included the sample code, type of container, number of organisms in the container, and a rough classification of various levels (e.g. bivalve, gastropod, crustacean, or recognizable common names such as crab, brittle star, etc.).
- In the process of logging, we identified which containers had more than one organism (labeled Bulk), duplicate sample codes, and containers without proper codes (format: WPS-000).
- Sorted the containers by ascending sample code in the shelves, and separated the bulk.
- Replaced the sample codes of all jars to match the proper format.
- Made a list of all sample codes present on a certain shelf (including cross-checking the physical containers with our database), printed the list, and posted it on the shelf.
- Samples that were not labeled correctly initially were not included and instead separated.

Insights and thoughts:

Handling large (I'm not sure if 400+ specimen is considered large relative to other projects) collections is very tedious but character-building. The repetitive process is easy to mess up, especially when logging data so double-checking everything digitally and physically was necessary. It was also very interesting to observe the variety of different invertebrates ranging from crabs to sea stars. Being so proximate to real samples is a privilege because there are so many research projects out there that need massive samples that aren't usually accessible to high schoolers.

This task helped me see the background work that researchers do outside of just research, and it made me appreciate biology a lot more – because even if it is tedious, biologists focus on attention to detail a lot more than physicists or mathematicians.

I used to cold-email a lot of professors asking if I could help at their labs for this exact purpose – to experience the menial work, but also to learn and ask lots of questions – and I finally got to experience it here.





What we did: Identified Pag-asa bulk samples

- Sorted, rehoused, labeled, and identified samples from the containers with more than one specimen. Specimen of the same species were put into the same containers
- Logged the information onto a new database.

What I did: Exploration of geometric morphometrics

- Read literature on the process and applications of geometric morphometrics (textbook [Geometric Morphometrics for Biologists: A Primer by Zelditch et al.], research papers).
- Set up the software to familiarize myself on how to do geometric morphometrics: R and RStudio using the geomorph package.
- Took photos of bivalves to test on.
- Tested out the placement of landmarks on several bivalves using the photos taken.

Insights and thoughts:

It was good that I was able to contribute a little bit to the documentation of Pag-asa island's biodiversity by identifying specimen that haven't been logged yet in a database. It was a challenge to identify some of the smaller shells since not all of the morphological features can be seen.

Regarding geometric morphometrics, I'm glad I got to get used to placing landmarks on homologous traits using real samples instead of online photos, because I got to familiarize myself also with taking photos. I'm still not clear on my direction with this method but I know it is a valuable process to learn because it's inevitable that I will have to use GM in the future for a study (since it is a rising method for taxonomy). It was also a refresher for using R and RStudio (not really because I was using a new package [geomorph] instead of what I did in statistics class, but still valuable).

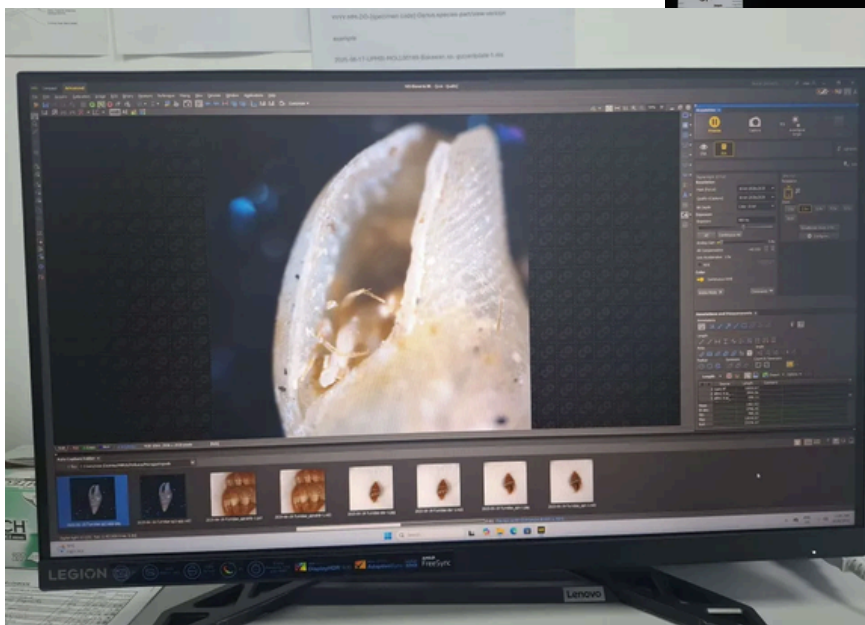
What I did: Photography of shells and research

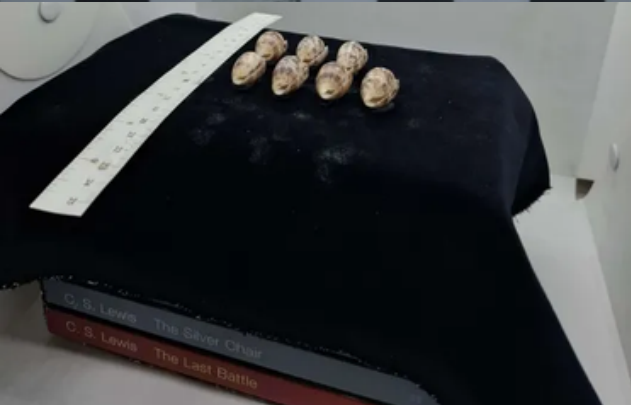
- Familiarized myself with the *Conus* samples belonging to MBRIS.
- Experimented with different camera settings and photo setups to photograph the shells (I initially borrowed a camera from Ma'am Machel, but in the succeeding days I brought my own digital camera).
- Took around 10-15 photos for each species with a black background (needed to redo it to add a scale).
- Read, and read, and read... many papers about classification, geometric morphometrics, and other miscellaneous stuff.
- Tried out the microscope for taking photos of micro snails! :)



Insights and thoughts:

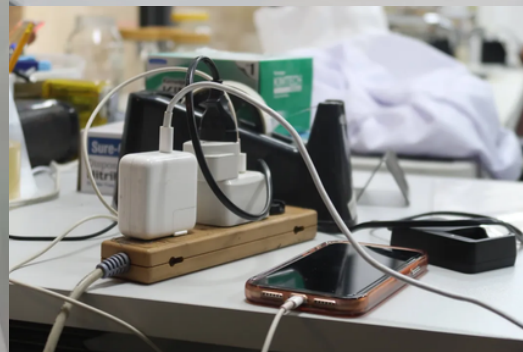
Classification is such a complicated and wide method especially for invertebrates. There are many ways to approach it even for seemingly simple specimen like *Conus*. The paper Ma'am Machel sent me even reinforces this because someone was able to analyze their shell shape using mathematical equations. Though methods have evolved to be more digital and advanced (e.g. GM, molecular methods such as DNA barcoding) there is incredible value in having the skills to classify specimen, especially those that are as diverse as invertebrates, visually – especially since all these more advanced methods need basis. I've also come to realize why data is the currency of science – it's very time consuming to collect.





What I did: Working more on the geometric morphometrics project!

- Changed my setup for taking photos so that the shells are closer to the camera for better resolution.
- Tried out ML-morph for automating geometric morphometrics on sample images.
- Presented my idea with a slideshow to everyone during a meeting.
- I didn't really do much technically... just had to read a lot to prepare for the presentation – especially the methods since they're all very new.
- I photographed over 300 specimen!



Insights and thoughts:

Data collection is still proving to be very tedious! I was able to create a more efficient method, but I am dreading the time when I will have to landmark all of these images (if ML-morph fails me). I don't have a lot to say about what I'm doing since it's mostly repetitive, and I'm not sure if this is something I'll do as my research project for school. I'm exploring other biophysics topics as well. It helps that I know the types of samples that are collected by MSI.

This week I ate lunch several times with the other interns and also talked more during work times. It was very interesting to hear about their college experiences and also what they're doing during the internship.



What I did: Final Presentation

- I finished data gathering (taking photos) on Tuesday so for the rest of the week I prepared for my final presentation for Dr. Malay.
- Edited past slides and did more research on the relevant methods.
- Experimented with different landmark placing softwares – tried R (geomorph), tpsdig, and imglab. My conclusion is that imglab is the most effective for my project.
- After the presentation, I analyzed sea cucumber ossicles using a microscope.
- I also prepared some microscope slides with scrapings of sea cucumbers.
- Read more papers about different types of shells, including the one Dr. Malay recommended to me.

Insights and thoughts:

I am so relieved to be done with data gathering! This means that even if my internship ended, I'm free to experiment with the photos. I wonder what other projects I could do with them other than the one I'm currently working on. Speaking of that, I started placing landmarks on some specimen. I might take a break for a while before continuing the project after this week is over, but I told myself I would work on it during the school year.

I think the presentation went well – it was much more polished than the one I did last week. I also got some feedback from Dr. Malay and everyone asked really good questions that also had me pondering. I hope people in marine science could find my project useful or interesting! For the last day, I just shadowed the other interns while they worked on the sea cucumbers. It was very interesting to find out that sea cucumbers can be identified by just their ossicles, which are microscopic. I wonder what causes the different shapes. There is also a large variety of sea cucumber species! I might look more into sea cucumbers in the future.





As a sort of conclusion, I am really grateful that I got this opportunity. Not everyone at my school gets to spend 200+ hours at an institute like MSI and be able to learn, ask questions, and be exposed to the projects done by researchers. I am really glad I decided to reach out to Dr. Malay and ask if I could stay here over the summer because it's been a wonderful experience. I learned a lot and finally got some direction for what I want to do for the next years. Marine science is the way to go (I'm still not sure what to do for undergrad though...)!! Everyone I met at this lab was amazing and super friendly, I hope to stay in touch with them even after this summer. Thank you so much to the other interns for being so nice, and Sir Claude, Ma'am Gizelle, and Dr. Malay for guiding me.

